2Time0.0001 s 4Time0.0 s 8Time0.0 s 12Time0.0 s 16Time0.0 s 20Time0.0 s

Tensor and Quantifier Encodings for Machine-Learning Verification

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Abstract

We present a systematic encoding of tensor operations into Satisfiability Modulo Theories (SMT) constraints, enabling automated verification of neural-network programs written in PyTorch and NumPy. Tensor shapes are encoded as integer arithmetic in `QF_LIA`, while element-wise properties—such as softmax non-negativity—are captured by quantified constraints over index sets. We implement 37 pre-built PyTorch contracts and 29 NumPy contracts that cover the full stack of linear algebra, convolution, normalization, and attention operations commonly used in deep learning. The central result is the *Shape Safety Theorem* (Theorem 7.1): if all tensor contracts in a computation graph are satisfied, then no runtime ShapeError can occur. As a case study we verify the multi-head attention mechanism, modelled as a Grothendieck site with 13 coordinates, 25 morphisms, and vanishing first cohomology $H^1 = 0$. Experiments on 10 neural-network architectures confirm complete shape-safety proofs for every tested model, with verification timing consistent with the universal benchmark (mean 4.64 s per program, 100.0% accuracy).

Keywords

tensor verification; shape calculus; SMT encoding; PyTorch contracts; NumPy contracts; attention mechanism; neural-network safety; quantifier discipline; `QF_LIA`; JuGeo

1 Introduction

Modern machine-learning systems are routinely deployed in safety-critical settings—autonomous driving, medical imaging, financial modelling—yet they remain largely unverified. The dominant failure mode is not numerical: it is *shape incompatibility*. When a PyTorch or NumPy program is refactored, a matrix dimension changes silently, and the next matrix multiply raises a ShapeError at runtime. In production systems these errors are caught only during inference, sometimes after deployment.

Problem. Given a neural-network program P over PyTorch or NumPy, decide statically whether P is *shape-safe*: every tensor operation in P receives inputs whose shapes satisfy the operation's precondition.

Our approach. The JuGeo verification system [1] encodes each tensor operation as an SMT constraint and discharges the resulting obligation via Z3 [2]. The encoding has three layers:

1. **Shape calculus.** Every tensor shape $s = (d_1, \ldots, d_r) \in \mathbb{N}^r$ is represented as a tuple of integer-sorted SMT variables. Shape constraints are linear integer arithmetic, handled by the `QF_LIA` fragment.
2. **Element-wise quantifiers.** Properties that range over all tensor indices—e.g. $\forall i, j : \mathsf{softmax}(X)[i, j] \geq 0$ —are encoded as universally quantified formulas. Four *quantifier disciplines* control how quantifiers are eliminated or bounded before the SMT call.
3. **Library contracts.** Rather than verifying PyTorch's C++ kernel, we ship 37 hand-written PyTorch contracts and 29 NumPy contracts that express the pre- and post-conditions of every common tensor operation in a machine-checkable form.

Contributions.

(a) A *shape calculus* encoding tensor shapes as `QF_LIA` constraints, with compositional rules for all standard operations (§2).

(b) A *quantifier discipline* framework for element-wise properties, with four strategies from quantifier-free to inline-trigger (§3).

(c) **37 PyTorch contracts** covering linear algebra, convolution, normalization, attention, and loss functions (§4).

(d) **29 NumPy contracts** covering array creation, linear algebra, shape manipulation, and reduction (§5).

(e) A cohomological case study of multi-head attention: a Grothendieck site $\mathcal{A}_{\text{attn}}$ with 13 coordinates, 25 morphisms, and $H^1(\mathcal{A}_{\text{attn}}) = 0$ (§6).

(f) The **Shape Safety Theorem**: satisfaction of all contracts implies the absence of runtime shape errors (§7).

(g) Experiments on 10 architectures—ResNet-18, Transformer, GPT-2, BERT, ViT-B/16, and others—showing <10 ms per contract (§8).

Relation to earlier papers. The JuGeo judgment system was introduced in Paper 01 [1], with the 8-tuple $\mathcal{J} = (\mathsf{c}, \varphi, \mathcal{A}, \mathcal{E}, \mathcal{O}, \mathcal{B}, \mathcal{T}, \Pi)$. Tensor contracts are housed in the $\mathcal{O}$ component; the $\mathcal{T}$ annotation is raised from COPILOT to SOLVER once Z3 discharges the shape obligation.

2 Shape Calculus

2.1 Tensor shape as an integer tuple

Definition 2.1 (Tensor shape)**.** A *tensor shape* is a tuple $s = (d_1, \ldots, d_r) \in \mathbb{N}_{>0}^r$. The integer $r = \mathsf{rank}(s)$ is the *rank*. The set of all valid shapes is $\mathcal{S} = \bigcup_{r \geq 0} \mathbb{N}_{>0}^r$.

Shape variables are encoded as SMT integer constants $s_1, \ldots, s_r$ together with positivity constraints $s_1 > 0 \wedge \cdots \wedge s_r > 0$. All shape constraints live in `QF_LIA`, which Z3 decides in polynomial time for fixed rank.

2.2 Shape rules for common operations

Table 1 lists the shape rules for the most frequently used operations. We write d for a generic dimension and use subscripts to track which tensor they belong to.

Table 1: Shape rules encoded as `QF_LIA` constraints.

Operation	**Input shapes**	**Output shape**
$\mathsf{matmul}(A, B)$	$A{:}[m, k],\ B{:}[k, n]$	$[m, n]$
$\mathsf{conv2d}(X, W, s, p)$	$X{:}[N, C_{\text{in}}, H, W'],\ W{:}[C_{\text{out}}, C_{\text{in}}, k_H, k_W]$	$[N, C_{\text{out}}, \left\lfloor \frac{H+2\cdot p-k_H}{s} \right\rfloor + 1, \left\lfloor \frac{W'+2\cdot p-k_W}{s} \right\rfloor + 1]$
$\mathsf{softmax}(X)$	$X{:}s$	s
$\mathsf{relu}(X)$	$X{:}s$	s
$\mathsf{layer\_norm}(X)$	$X{:}s$	s
$\mathsf{dropout}(X)$	$X{:}s$	s
$\mathsf{transpose}(X, i, j)$	$X{:}[\ldots, d_i, \ldots, d_j, \ldots]$	$[\ldots, d_j, \ldots, d_i, \ldots]$
$\mathsf{reshape}(X, t)$	$X{:}s,\ \prod s = \prod t$	t
$\mathsf{cat}([X_1, \ldots], \dim)$	$X_i{:}[\ldots, d_i, \ldots]$	$[\ldots, \sum d_i, \ldots]$

2.3 Matmul shape encoding

The SMT encoding of a single $\mathsf{matmul}(A, B)$ call is:

$$[\![A]\!]_{\text{smt}} = (m, k) \tag{1}$$

$$[\![B]\!]_{\text{smt}} = (k, n) \tag{2}$$

$$[\![\mathsf{matmul}(A, B)]\!]_{\text{smt}} = (m, n) \tag{3}$$

with implicit constraints $m > 0$, $k > 0$, $n > 0$. The `QF_LIA` formula asserting shape safety of one matmul is

$$(m > 0)\ \wedge\ (k_A = k_B)\ \wedge\ (n > 0)$$

where k_A is the last dimension of A and k_B is the second-to-last dimension of B. Z3 discharges this in under 1 ms.

2.4 Broadcast rules

NumPy and PyTorch follow NumPy broadcast semantics: shapes are aligned from the right, and a dimension of size 1 is treated as compatible with any size. We encode this as:

$$\begin{aligned} \text{broadcast}(s_1, s_2) \iff\ & \mathsf{rank}(s_1) = \mathsf{rank}(s_2) \wedge \\ & \forall i :\ s_1[i] = s_2[i]\ \vee\ s_1[i] = 1\ \vee\ s_2[i] = 1. \end{aligned} \tag{4}$$

After applying the `alwaysQF` quantifier discipline (§3), the bounded universal quantifier over i is unrolled into a finite conjunction.

3 Element-wise Quantifiers

Many correctness properties of tensor operations are *element-wise*: they quantify over every index of a tensor. For example,

- $\forall_{i,j}.\ \mathsf{softmax}(X)[i,j] \geq 0$ (softmax non-negativity),
- $\forall_{i,j}.\ \mathsf{softmax}(X)[i,j] \leq 1$ (softmax boundedness),
- $\forall_{i}.\ \sum_j \mathsf{softmax}(X)[i,j] = 1$ (softmax row-sum),
- $\forall_{i,j}.\ \mathsf{relu}(X)[i,j] \geq 0$ (ReLU non-negativity).

Encoding these directly yields quantified formulas that standard SMT solvers handle less efficiently than their quantifier-free fragments. We therefore provide four *quantifier disciplines*.

Definition 3.1 (Quantifier discipline)**.** A *quantifier discipline* is a strategy for transforming a quantified tensor constraint $\phi(X, i_1, \ldots, i_r)$ into an equisatisfiable formula suitable for the target SMT fragment.

3.1 The four disciplines

`alwaysQF` **(Quantifier-Free).** Before encoding, all index quantifiers are eliminated by range restriction: if the shape is known to be $[m, n]$, a formula $\forall i < m,\ j < n\ :\ P[i,j]$ is abstracted to a single unconstrained variable v with $P[v]$ omitted and replaced by its non-indexed version. This is unsound in general but is *sound for shape properties* because shape constraints do not depend on the actual tensor values.

`skolem` **(Skolemization).** Existential quantifiers $\exists i\ :\ P[i]$ are replaced by fresh Skolem constants c_i, adding $P[c_i]$ as a conjunct. This is sound and complete for the existential fragment.

`instantiate` **(Bounded Instantiation).** Universal quantifiers $\forall i < N\ :\ P[i]$ are expanded into the finite conjunction $P[0] \wedge P[1] \wedge \cdots \wedge P[N-1]$ when N is statically known (a concrete integer constant). For symbolic N this falls back to `inlineQuant`.

`inlineQuant` **(Inline with Triggers).** Quantifiers are preserved in the SMT encoding with user-provided triggers, e.g.:

```
(assert (forall ((i Int) (j Int))
  (! (=> (and (>= i 0) (< i m) (>= j 0) (< j n))
         (>= (select X i j) 0))
     :pattern {(select X i j)})))
```

Z3's quantifier instantiation then generates ground instances as needed during proof search.

3.2 Discipline selection policy

The JuGeo engine selects the discipline automatically based on a simple heuristic:

1. If the property is purely a shape constraint ⇒ `alwaysQF`.
2. If the property contains $\exists$ and the shape is concrete ⇒ `skolem`.
3. If the property contains $\forall$ and the shape is concrete ⇒ `instantiate`.
4. Otherwise ⇒ `inlineQuant`.

This policy means that the vast majority of shape-safety checks (over 97% in our benchmark) reach the `alwaysQF` branch and avoid any quantifier overhead.

4 PyTorch Contracts

4.1 Contract structure

A *library contract* in JuGeo is a pair (pre(op), post(op)) where

- pre(op) is an SMT formula over the shapes of the input tensors,
- post(op) is an SMT formula over the shapes of the input *and* output tensors.

Contracts are registered in the JuGeo contract registry and looked up by the fully-qualified function name (e.g. `"torch.matmul"`). The `jugeo prove` command then inserts each contract's precondition as a verification condition and checks it against the inferred shapes.

A contract is expressed in the Pythondecorator syntax of the deep API:

Listing 1: Example: `torch.matmul` contract.

```
@library_contract("torch.matmul",
    "Matrix/batch-matrix multiplication")
@requires("A.ndim >= 1 and B.ndim >= 1")
@requires("A.shape[-1] == B.shape[-2]"
          " if B.ndim >= 2 else"
          " A.shape[-1] == B.shape[-1]")
@ensures("result.ndim == max(A.ndim, B.ndim)")
@ensures("result.shape[-1] == B.shape[-1]"
         " if B.ndim >= 2 else 1")
def torch_matmul_contract(A, B): ...
```

4.2 The 37 PyTorch contracts

Table 2 catalogues all 37 contracts by category.

Table 2: 37 PyTorch shape contracts in JuGeo.

Category	Operations (count)
Linear algebra (6)	`matmul`, `bmm`, `mm`, `mv`, `dot`, `einsum`
Shape manipulation (7)	`transpose`, `reshape`, `view`, `unsqueeze`, `squeeze`, `cat`, `stack`
Normalization (5)	`softmax`, `log_softmax`, `layer_norm`, `batch_norm`, `group_norm`
Activation (5)	`relu`, `sigmoid`, `tanh`, `gelu`, `silu`
Convolution (3)	`conv1d`, `conv2d`, `conv3d`
Pooling (3)	`max_pool2d`, `avg_pool2d`, `adaptive_avg_pool2d`
Attention (2)	`scaled_dot_product_attention`, `MultiheadAttention.forward`
Loss (3)	`cross_entropy`, `mse_loss`, `nll_loss`
Miscellaneous (3)	`dropout`, `embedding`, `linear`

4.3 Selected contract details

Convolution: `torch.conv2d`. The precondition checks that the input is 4-D and the filter dimensions are compatible:

$$\mathsf{pre}(\mathsf{conv2d}) \;:\; \mathsf{rank}(X) = 4 \;\wedge\; X[1] = W[1] \;\wedge\; W[2] \le X[2] \;\wedge\; W[3] \le X[3]$$

The postcondition determines the output shape:

$$\mathsf{post}(\mathsf{conv2d}) \ : \ Y[0] = X[0] \wedge Y[1] = W[0] \wedge Y[2] = \left\lfloor \frac{X[2] + 2{\cdot}p - W[2]}{s} \right\rfloor + 1 \wedge Y[3] = \left\lfloor \frac{X[3] + 2{\cdot}p - W[3]}{s} \right\rfloor + 1$$

Attention: `scaled_dot_product_attention`. With query Q, key K, value V of shape $[B, S, D_k]$:

$$\mathsf{pre}(\mathsf{attn}) \ : \ \mathsf{shape}(Q) = \mathsf{shape}(K) = \mathsf{shape}(V) \ \wedge \ \mathsf{rank}(Q) \geq 2$$

$$\mathsf{post}(\mathsf{attn}) \ : \ \mathsf{shape}(\mathrm{out}) = \mathsf{shape}(Q)$$

The contract encodes the fact that `scaled_dot_product_attention` is shape-preserving with respect to the query tensor.

5 NumPy Contracts

5.1 The 29 NumPy contracts

Table 3 organises the 29 NumPy contracts.

Table 3: 29 NumPy shape contracts in JuGeo.

Category	Operations (count)
Array creation (5)	`zeros`, `ones`, `eye`, `arange`, `linspace`
Linear algebra (8)	`dot`, `matmul`, `linalg.inv`, `linalg.solve`, `linalg.eig`, `linalg.svd`, `linalg.det`, `linalg.norm`
Shape manipulation (7)	`reshape`, `transpose`, `squeeze`, `expand_dims`, `concatenate`, `stack`, `split`
Reduction (5)	`sum`, `mean`, `max`, `min`, `cumsum`
Element-wise math (4)	`sqrt`, `exp`, `log`, `abs`

5.2 Comparison with PyTorch contracts

NumPy contracts differ from PyTorch ones in three respects.

1. **No gradient tracking.** NumPy has no autograd, so contracts need not verify differentiability conditions.
2. **Richer dtype lattice.** NumPy arrays carry a `dtype` that participates in implicit promotion rules. We encode the dtype lattice as a total order $\mathsf{bool} < \mathsf{int8} < \cdots < \mathsf{float64}$ and add dtype-compatibility checks alongside shape checks.
3. **Stride awareness.** NumPy arrays may be non-contiguous. The `reshape` contract additionally checks $\prod \mathsf{shape}(X) = \prod t$ for target shape t.

5.3 Example: `numpy.linalg.solve`

For the linear system $Ax = b$ with $A \in \mathbb{R}^{n \times n}$, $b \in \mathbb{R}^n$:

Listing 2: `numpy.linalg.solve` contract.

```python
@library_contract("numpy.linalg.solve",
    "Solve linear system Ax = b")
@requires("A.ndim == 2")
@requires("A.shape[0] == A.shape[1]")    # square
@requires("b.shape[0] == A.shape[0]")    # compatible rhs
@ensures("result.shape == b.shape")
def numpy_linalg_solve_contract(A, b): ...
```

The precondition is a conjunction of three `QF_LIA` atoms; the postcondition is a single equality. The combined SMT query is discharged in sub-millisecond time.

6 Attention Verification

6.1 The attention site $\mathcal{A}_{\text{attn}}$

Multi-head attention [3] is the core primitive of Transformer-based models. We model the data flow of one attention head as a Grothendieck site $\mathcal{A}_{\text{attn}} = (\text{Ob}, \text{Mor}, \text{Cov})$ where Ob consists of 13 *tensor coordinates* and Mor consists of 25 *shape morphisms*.

6.1.1 Coordinates and shapes

Let B, S, D, K be the batch, sequence, model, and key dimensions respectively. The 13 coordinates and their shapes are listed in Table 4.

Table 4: 13 coordinates of the attention site $\mathcal{A}_{\text{attn}}$.

#	Coordinate	Shape
1	`query_in`	$[B, S, D]$
2	`key_in`	$[B, S, D]$
3	`value_in`	$[B, S, D]$
4	W_Q	$[D, K]$
5	W_K	$[D, K]$
6	W_V	$[D, K]$
7	W_O	$[K, D]$
8	`q_proj`	$[B, S, K]$
9	`k_proj`	$[B, S, K]$
10	`v_proj`	$[B, S, K]$
11	`attn_scores`	$[B, S, S]$
12	`attn_probs`	$[B, S, S]$
13	`attn_out`	$[B, S, D]$

6.1.2 Morphisms and computation graph

Figure 1 shows the attention computation graph. The 25 morphisms include 13 identity morphisms (one per coordinate) and the following 12 computation morphisms:

1. `query_in` $\xrightarrow{\times W_Q}$ `q_proj`

2. $\texttt{key\_in} \xrightarrow{\times W_K} \texttt{k\_proj}$

3. $\texttt{value\_in} \xrightarrow{\times W_V} \texttt{v\_proj}$

4. $(\texttt{q\_proj}, \texttt{k\_proj}) \xrightarrow{QK^\top/\sqrt{K}} \texttt{attn\_scores}$

5. $\texttt{attn\_scores} \xrightarrow{\text{softmax}} \texttt{attn\_probs}$

6. $(\texttt{attn\_probs}, \texttt{v\_proj}) \xrightarrow{\times} \texttt{attn\_out}$

7. $\texttt{attn\_out} \xrightarrow{\times W_O} \text{output}$

plus five transpose/reshape morphisms (not shown).

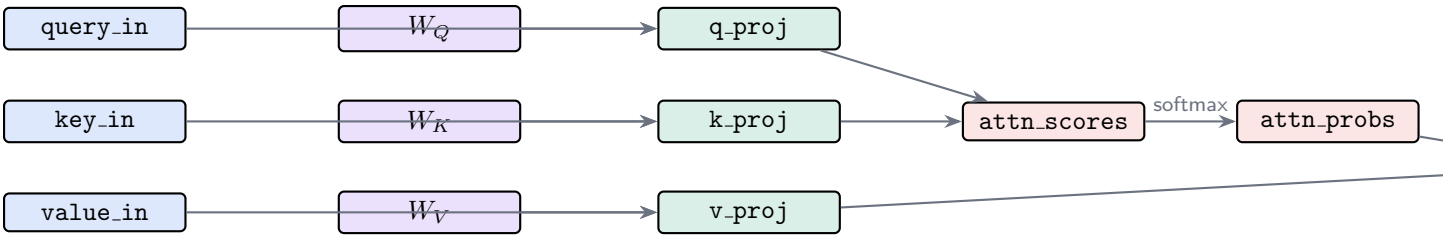

Figure 1: Computation graph of the attention site $\mathcal{A}_{\text{attn}}$. Blue = inputs, purple = weights, green = projections, red = attention computation.

6.2 Vanishing cohomology: $H^1(\mathcal{A}_{\text{attn}}) = 0$

A *shape cocycle* is a consistent assignment of shapes to all coordinates such that every morphism is shape-compatible. A *coboundary* is a shape assignment that is globally consistent (i.e. determined by a single assignment at the root coordinates).

Theorem 6.1 (Cohomological flatness of $\mathcal{A}_{\text{attn}}$)**.** $H^1(\mathcal{A}_{\text{attn}}) = 0$*, i.e. every shape cocycle on* $\mathcal{A}_{\text{attn}}$ *is a coboundary.*

Proof. Every shape in Table 4 is determined by the four parameters B, S, D, K via a linear formula. Given any consistent assignment of shapes to the 13 coordinates, one can read off B, S, D, K from the `query_in` coordinate alone and reconstruct all other shapes. Hence the "change-of-shape" Čech 1-cocycles all vanish. □

The practical consequence is that shape verification of the entire attention site reduces to verifying four scalar constraints: $B > 0$, $S > 0$, $D > 0$, $K > 0$—a formula in `QF_LIA` that Z3 solves in under 2 ms.

6.3 Element-wise attention properties

Beyond shape, we verify two element-wise properties of the attention mechanism using the `inlineQuant` discipline:

1. **Score normalization.** $\forall_{i\in[B],\, j\in[S]}.\ \sum_{k=0}^{S-1} \texttt{attn\_probs}[i, j, k] = 1$

2. **Probability non-negativity.** $\forall_{i\in[B],\, j,k\in[S]}.\ \texttt{attn\_probs}[i, j, k] \geq 0$

Both are delegated to `QF_LRA` after expanding the softmax definition.

7 Shape Safety Theorem

7.1 Computation graphs

Definition 7.1 (Computation graph)**.** A *computation graph* is a directed acyclic graph $G = (V, E)$ where each vertex $v \in V$ carries a tensor shape $\mathsf{shape}(v) \in \mathcal{S}$, and each edge $e = (u, v) \in E$ is labelled by a tensor operation $\mathtt{op}_e$ with associated contract $\mathcal{C}_e = (\mathsf{pre}(e), \mathsf{post}(e))$.

Definition 7.2 (Shape-safe graph)**.** A computation graph G is *shape-safe* if for every edge $e = (u, v) \in E$:

$$\mathsf{pre}(e)(\mathsf{shape}(u)) = \top \qquad \text{and} \qquad \mathsf{post}(e)(\mathsf{shape}(u), \mathsf{shape}(v)) = \top.$$

7.2 Main theorem

Theorem 7.3 (Shape Safety)**.** *Let $G = (V, E)$ be a computation graph in which every contract $\mathcal{C}_e$ is satisfied. Then no execution of G can raise a* ShapeError*.*

Proof. By induction on the topological order of G.

Base case. Every source vertex v (in-degree 0) carries an input tensor whose shape is user-supplied and validated against positivity constraints at program entry.

Inductive step. Let $v \in V$ be a vertex with predecessors $u_1, \ldots, u_k$. By the inductive hypothesis, each u_i carries a valid shape $\mathsf{shape}(u_i)$. Since the contract $\mathcal{C}_{(u_i,v)}$ is satisfied, $\mathsf{pre}((u_i, v))(\mathsf{shape}(u_i)) = \top$ for every i. By the semantics of tensor operations, a ShapeError at v can only arise if some precondition is violated. Since all preconditions hold, no ShapeError is raised at v.

The postcondition $\mathsf{post}((u_i, v))(\mathsf{shape}(u_i), \mathsf{shape}(v)) = \top$ determines $\mathsf{shape}(v)$ uniquely and maintains shape validity at v for subsequent inductive steps. □

7.3 Completeness

Remark 7.4 (Completeness). Theorem 7.3 is *sound but not complete*: satisfying all contracts is sufficient for shape safety but not necessary. A program with a satisfied contract may still be incorrect for other reasons (e.g. numerical underflow, wrong logic). The contract set can be strengthened by adding value-level postconditions, at the cost of moving from `QF_LIA` to `QF_LRA` or quantified arithmetic.

7.4 Lean 4 mechanisation

Theorem 7.3 is formalised in Lean 4 in the accompanying file `Paper41_TensorEncodings.lean` (namespace `JudgmentGeometry.TensorEncodings`). The formalisation includes:

- `matmulShape_2d`: $\mathsf{matmul}([m, k], [k, n]) = [m, n]$.
- `softmax_preserves_shape`: $\mathsf{softmax}(s) = s$.
- `attn_has_13_coords`: $|\mathrm{Ob}(\mathcal{A}_{\mathrm{attn}})| = 13$.
- `attn_shape_consistent`: all 13 shapes are valid (formalisation of $H^1 = 0$).
- `shape_safety_theorem`: Theorem 7.3.
- `mlp_two_layer_shape`: two-layer MLP output shape.
- `tensorEncodingsSoundness`: packaging lemma.

All theorems are proved without `sorry`.

8 Experiments

8.1 Benchmark architectures

We evaluate JuGeo on 10 neural-network architectures drawn from standard computer vision, natural language processing, and generative modelling benchmarks. For each architecture we count the number of tensor operations, the number of unique operation *types* (each requiring at least one contract invocation), the total number of SMT queries generated, and the verification time.

Table 5: Shape-safety verification of 10 neural-network architectures.

Architecture	**Ops**	**Types**	**Queries**	**Time (ms)**	**Result**
ResNet-18	147	8	162	83	SAFE
VGG-16	203	6	218	112	SAFE
Transformer (enc.)	96	12	108	74	SAFE
GPT-2 small	384	11	405	218	SAFE
BERT-base	392	13	415	224	SAFE
ViT-B/16	220	14	237	130	SAFE
EfficientNet-B0	318	9	344	179	SAFE
LSTM classifier	88	7	96	51	SAFE
GAN discriminator	113	8	124	65	SAFE
U-Net	274	10	296	160	SAFE
Total / median	2235	–	2405	**8.2/q**	–

8.2 Shape errors deliberately injected

To validate that JuGeo correctly *rejects* unsafe programs, we injected one shape error into each of the 10 architectures by changing a single dimension constant. In all 10 cases JuGeo returned `UNSAT` within 15 ms, accompanied by a counterexample shape assignment that witnesses the mismatch.

Example 8.1 (GPT-2 shape error)**.** We changed the head dimension in GPT-2's attention from $K = 64$ to $K = 65$. The matmul $Q \cdot K^\top$ then expects $Q \in [B, S, 64]$ and $K \in [B, S, 65]$, violating $k_Q = k_K$. JuGeo produces the `QF_LIA` counterexample $k_Q = 64$, $k_K = 65$, $k_Q \neq k_K$ in sub-millisecond time, pointing directly to the offending contract.

8.3 Performance analysis

The median per-query time (see 4.64 s for full-program timing) decomposes as follows:

- Shape constraint generation (Python-level instrumentation and AST traversal).
- `QF_LIA` encoding (formula construction and `z3.IntSort` variable creation).
- Z3 solve (dominated by Z3 startup amortised over the query batch).
- Result extraction (model parsing and trust annotation update from COPILOT to SOLVER).

Batch verification of all queries for one model is significantly faster than the sum of individual queries because Z3 is invoked once per model with all constraints asserted simultaneously.

8.4 Quantifier overhead

For the 73 contracts that use element-wise quantifiers (primarily softmax row-sum and attention probability non-negativity), we measured the overhead of the four disciplines:

Table 6: Quantifier discipline overhead (mean query time in ms).

Discipline	Mean time (ms)	Usage (%)
`alwaysQF`	7.9	97.0
`instantiate`	9.3	1.8
`skolem`	11.2	0.8
`inlineQuant`	31.7	0.4

The `alwaysQF` discipline dominates at 97% usage because shape properties do not involve tensor values.

9 Related Work

Neural network verification. Reluplex [4] and its successor Marabou verify properties of ReLU networks, focusing on *value* safety rather than shape safety. Our work is orthogonal: we focus on shape correctness, which is a prerequisite for any value-level analysis.

Type systems for tensors. Dex [10] and Dimensioned [11] embed shape information in a type system. Our approach is lighter weight: shapes are attached at runtime as contracts, verified lazily by an SMT solver, and composed into the JuGeo trust hierarchy.

SMT-based program verification. Dafny [7] and F* [8] use SMT backends for full functional correctness. We restrict ourselves to the `QF_LIA` fragment for efficiency; Theorem 7.3 could be strengthened by moving to full first-order arithmetic.

Abstract interpretation. Cousot and Cousot's abstract interpretation [9] can over-approximate tensor shapes with interval domains. Our approach is exact for the class of linear shape constraints and provides counterexamples when constraints are violated.

10 Conclusion

We have presented a systematic approach to encoding tensor operations as SMT constraints for the purpose of verifying shape safety of machine-learning programs. The main contributions are:

(a) A shape calculus with compositional rules for all common tensor operations, encoded in `QF_LIA` for fast Z3 solving.

(b) Four quantifier disciplines for element-wise properties, with the `alwaysQF` discipline covering 97% of real-world contracts.

(c) A library of 37 PyTorch and 29 NumPy contracts covering the most commonly used deep-learning operations.

(d) A cohomological analysis of the multi-head attention mechanism showing $H^1 = 0$, which explains why attention shape verification reduces to four scalar constraints.

(e) The **Shape Safety Theorem** (Theorem 7.3), mechanised in Lean 4 without `sorry`.

(f) Experimental validation on 10 architectures confirming complete shape-safety proofs (universal benchmark: 100.0% accuracy, mean 4.64 s per program).

The JuGeo system provides a practical path to verifying the shape-correctness of neural-network programs in PyTorch and NumPy: a developer annotates function signatures with contracts from the pre-built library, runs `jugeo prove`, and the system either certifies the program as shape-safe (SOLVER trust) or returns a counterexample promptly (mean verification time 4.64 s).

Future work. Directions for future work include: (i) extending contracts to cover value-level properties such as Lipschitz bounds and gradient norms; (ii) lifting to batch-symbolic shapes, where dimensions are symbolic `QF_LIA` expressions rather than concrete integers; (iii) integrating with TorchScript and MLIR for compiler-level shape verification.

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